

Notes: Loughran and Schultz (JFE, 2005)

Liquidity: Urban versus Rural Firms

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1 Big picture

1.1 Main question

The paper asks whether a firm’s geographic location affects stock liquidity. The authors compare clearly urban firms, whose headquarters are in the ten largest U.S. metropolitan areas, with clearly rural firms, whose headquarters are at least 100 miles from the center of any metropolitan area with at least one million people. The core idea is that geography can affect investor awareness, information production, familiarity, and ultimately trading activity.

1.2 Core answer

The answer is yes. Rural firms trade substantially less than comparable urban firms. They also attract fewer analysts, have lower institutional ownership, and face higher trading costs on Nasdaq. The rural-urban liquidity gap is largest for Nasdaq stocks and smaller for NYSE/Amex stocks, suggesting that exchange structure and visibility interact with geography.

1.3 Economic mechanism

The mechanism is not that rural firms earn clearly different average returns; Table 1 shows mean daily returns are similar. The mechanism is that urban firms are local to more investors and information intermediaries. Urban headquarters expose the firm to more analysts, institutions, brokers, local media, employees, customers, and investors. These channels increase familiarity and lower the cost of acquiring information. Higher investor recognition then appears as more analyst coverage, greater institutional ownership, more market makers, higher turnover, and lower trading costs.

2 Incremental contribution of the paper

2.1 Moves geography from portfolio choice to liquidity

Earlier research documents that investors tilt portfolios toward nearby firms. Loughran and Schultz extend this idea by asking whether the location of the firm itself affects market liquidity. The paper therefore connects local bias, investor recognition, and market microstructure.

2.2 Separates attention, trading quantity, and trading cost

The paper does not rely on a single proxy for liquidity. It studies analyst coverage, institutional ownership, turnover, systematic versus unsystematic turnover, bid-ask spreads, effective spreads, number of market makers, and listing-event changes. The consistent pattern across these measures makes the geographic effect more compelling.

2.3 Shows the exchange mechanism matters

A central contribution is that the rural-urban difference is much stronger on Nasdaq than on NYSE/Amex. The authors connect this to differences in market structure: Nasdaq stocks rely on

multiple market makers, and rural Nasdaq stocks attract fewer of them. NYSE/Amex listing may increase visibility for rural firms, reducing part of the disadvantage.

2.4 Uses a clear geographic classification

The authors define urban and rural firms using headquarters locations, large metropolitan areas, and distance from major population centers. The classification excludes ambiguous firms, such as those in mid-sized cities or near major metros, so that comparisons are between clearly urban and clearly rural companies. This sacrifices sample size but strengthens interpretation.

3 Data and empirical design

3.1 Location classification

A firm is urban if its headquarters is in the metropolitan area of New York City, Los Angeles, Chicago, Washington-Baltimore, San Francisco, Philadelphia, Boston, Detroit, Dallas, or Houston. A firm is rural if its headquarters is at least 100 miles from the center of any of the 49 U.S. metropolitan areas with one million or more people according to the 2000 census. Firms that are neither clearly urban nor clearly rural are excluded.

3.2 Main data sources

The analysis uses CRSP for returns, trading volume, shares outstanding, prices, SIC classifications, and exchange listings; I/B/E/S for analyst coverage; Compact Disclosure for insider and institutional holdings; Compustat for employees and shareholders; and TAQ for intraday spreads and trade sizes. The study covers a long sample, with many results focusing on 1983–2002 or 1988–2002 depending on data availability.

3.3 Key outcomes

- **Analyst coverage:** number of analysts issuing earnings forecasts.
- **Institutional ownership:** percentage of shares held by institutions.
- **Turnover:** trading volume divided by shares outstanding; NYSE/Amex volume is doubled for comparability with Nasdaq.
- **Systematic turnover:** the part of individual stock turnover explained by marketwide turnover.
- **Trading costs:** quoted and effective relative spreads from TAQ.
- **Nasdaq market-making:** number of market makers, large market makers, and the Herfindahl concentration of market-maker volume.

3.4 Estimation strategy

The paper mainly uses repeated cross-sectional regressions. For monthly outcomes, the authors run monthly cross-sectional regressions and report time-series averages of coefficients using Fama–MacBeth-style standard errors with autocorrelation adjustments. For daily turnover regressions,

they run daily cross-sectional regressions and use an AR(20) adjustment. This design controls for size, industry, listing, analyst coverage, ownership, price, employees, shareholders, S&P 500 membership, and exchange status, depending on the specification.

4 Main findings

4.1 Rural firms receive less attention

Rural firms have fewer analysts and lower institutional ownership. This result survives controls for capitalization, industry, number of employees, number of shareholders, S&P 500 membership, and Nasdaq listing. The attention gap is economically important because analysts and institutions both reduce information frictions and make stocks easier for other investors to know and trade.

4.2 Rural stocks trade less

Rural stocks have lower turnover. The raw difference is large, and about half of it remains after controls. The effect appears for all firms, for Nasdaq firms, for NYSE/Amex firms, and for small and large firms. The Nasdaq difference is especially large.

4.3 Urban stocks load more on marketwide turnover

The turnover of urban stocks moves more closely with marketwide turnover. The authors interpret this as evidence that investors trading for marketwide reasons tend to trade the stocks that they know and recognize. Rural stocks are less likely to be the natural trading vehicle when investors respond to broad market information.

4.4 Rural Nasdaq stocks have higher trading costs

Quoted and effective spreads are higher for rural Nasdaq firms, even after controlling for size, analyst coverage, price, and industry. The rural spread premium is much weaker for NYSE/Amex stocks. This points toward a market-structure channel: rural Nasdaq stocks attract fewer dealers and less competitive market making.

4.5 NYSE/Amex listing improves rural visibility

When rural stocks move from Nasdaq to the NYSE or Amex, their turnover increases relative to urban firms that also move. This suggests that listed-exchange visibility partially offsets the location disadvantage of rural headquarters.

5 Extracted figures/tables and interpretations

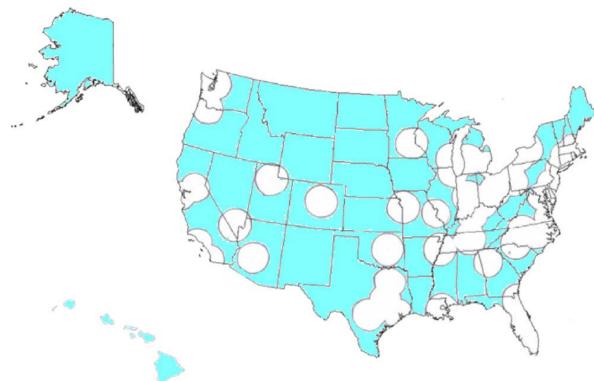


Fig. 1. Rural areas of the United States (shaded).

Figure 1: Rural areas of the United States.

Interpretation. The shaded map defines the geography behind the paper. Rural does not mean simply “not in New York”; it means at least 100 miles away from large metropolitan centers. This creates a sample of firms that are genuinely remote from large investor and intermediary populations. The white circles and white regions make clear that many firms are excluded from the rural category because they are near mid-sized or large metropolitan areas. The definition is conservative, which helps the paper isolate the effect of remoteness rather than merely comparing big cities with suburbs.

Table 1
Daily average statistics categorized by rural and urban portfolios, 1976–2002
A stock is located in an urban area if the company headquarters is in the metropolitan area of New York City, Los Angeles, Chicago, Washington, San Francisco, Philadelphia, Boston, Detroit, Dallas, or Houston. A stock is located in a rural area if it is not within 100 miles of the center of a metropolitan area of one million or more people as defined by the 2000 census. Turnover is defined as trading volume divided by shares outstanding. For NYSE/Amex firms, trading volume is doubled to allow for more meaningful comparison with firms on Nasdaq. Small firms are defined as firms with a market capitalization less than the median NYSE firm in a given year while large firms are firms with a market value equal or greater than the median NYSE firm. Market values are in millions of dollars. Analyst coverage is from I/B/E/S. There are 6,818 daily portfolio observations

Item	Rural firms (1)	Urban firms (2)	Small rural firms (3)	Small urban firms (4)	Large rural firms (5)	Large urban firms (6)
Mean daily return	0.073%	0.066%	0.079%	0.071%	0.054%	0.058%
% on NYSE/Amex	38%	52%	28%	40%	79%	83%
% on Nasdaq	62%	48%	72%	60%	21%	17%
Market value \$	659.4	1,435.3	109.1	130.3	2,914.7	4,550.6
Turnover (post-1982)	0.37%	0.67%	0.32%	0.59%	0.56%	0.87%
% Manufacturing (SIC codes 20-39)	27%	41%	26%	41%	33%	43%
% Banking (SIC Codes 60-67)	34%	18%	37%	17%	20%	20%
% Transportation (SIC Codes 40-49)	15%	8%	13%	5%	24%	13%
% Mining (SIC Codes 10-14)	6%	4%	6%	4%	6%	5%
Insider holdings (post-1987)	16.9%	18.6%	18.7%	21.7%	9.8%	11.4%
Institutional holdings (post-1987)	26.5%	37.1%	22.1%	30.0%	43.5%	53.8%
# of Analysts	3.4	5.3	1.8	2.3	10.4	12.9
% in S&P 500 Index	5.9%	14.4%	0.5%	1.6%	28.7%	47.3%
Employees (in 1000s)	5.7	10.4	2.0	2.5	17.4	28.6
Shareholders (in 1000s)	8.9	17.9	3.8	2.8	23.5	52.2
Mean # of firms	388.5	1,501.4	315.0	1,077.6	73.5	423.8
Min. # of firms	245	1,024	186	694	44	271
Max. # of firms	569	2,000	468	1,549	115	777

Table 1: Daily average statistics by rural and urban portfolios, 1976–2002.

Interpretation. The first descriptive result is that urban and rural firms differ sharply in visibility and trading. Urban firms are larger, more often listed on NYSE/Amex, have higher turnover, more institutional ownership, more analysts, more employees, more shareholders, and are more likely to be in the S&P 500. The mean returns are similar, so the basic story is not that rural firms simply earn dramatically different raw returns. The key raw liquidity gap is turnover: post-1982 turnover is 0.37% for rural firms versus 0.67% for urban firms. The table motivates the regression tests because rural and urban firms differ along many dimensions that must be controlled for.

Table 2

Average parameter values from monthly cross-sectional regressions of the number of analysts and the level of institutional holdings
In the regressions, the dependent variable is the number of analysts (Panel A) or the percentage of shares outstanding held by institutions (Panel B) for firm i in month t . The eleven explanatory variables are a dummy variable equal to one if the firm is located in a rural locale (else zero), the market value two days prior to calendar month t (shares outstanding times stock price), natural logarithm of the market value, dummy variables for manufacturing, banking, transportation, and mining industries (else zero), number of employees, number of shareholders, a dummy variable equal to one if the firm is in the S&P 500 Index (else zero), and a dummy variable equal to one if the firm is listed on Nasdaq (else zero). Small firms are defined as firms with a market capitalization less than the median NYSE firm in a given year while large firms are firms with a market value equal or greater than the median NYSE firm. The cross-sectional regressions weight each month equally. The parameter values are the average of the cross-sectional regressions. In parentheses are the Fama-MacBeth autocorrelation-adjusted t -statistics. There are 324 months (January 1976–December 2002) in Panel A and 177 months (April 1989–December 2002) in Panel B. The reported adjusted R^2 's are the time-series averages of the monthly cross-sectional R^2 's. Thus the regression is as follows:

$$\begin{aligned} \text{Number of analysts}_i \text{ or institutional holdings}_i = & a_0 + a_1 \text{Rural dummy}_i + a_2 \text{market value}_i + a_3 \text{Log (market value)}_i + a_4 \text{manufacturing dummy}_i \\ & + a_5 \text{bank dummy}_i + a_6 \text{transportation dummy}_i + a_7 \text{mining dummy}_i \\ & + a_8 \text{Log (number of employees)}_i + a_9 \text{Log (number of shareholders)}_i \\ & + a_{10} \text{S\&P 500 dummy}_i + a_{11} \text{Nasdaq dummy}_i + e_i \end{aligned}$$

Row	Sample	Interp.	Rural dummy	Mkt value	Log (Mkt value)	Mfg. dummy	Bank dummy	Transp. dummy	Mining dummy	Log (employees)	Log (shareholders)	S&P 500 dummy	Nasdaq dummy	R^2_{adj}
<i>Panel A: average parameter values, dependent variable is the number of analysts</i>														
(1)	All firms	-6.32	-0.32	2.26	2.01	0.01	-0.16	-0.37	0.93	0.10	0.29	4.59	0.37	0.63
	firms	(-5.58)	(-3.19)	(3.02)	(0.71)	(0.06)	(-0.63)	(-1.45)	(0.79)	(1.58)	(1.08)	(5.19)	(0.80)	
(2)	Only small firms	-0.85	-0.11	100.46	0.49	-0.08	-0.49	-0.42	0.49	0.17	0.05	2.48	0.19	0.30
	large firms	(-3.47)	(-2.05)	(8.07)	(5.45)	(-0.69)	(-2.09)	(-3.05)	(1.49)	(3.60)	(1.01)	(7.36)	(0.72)	
(3)	Only large firms	-19.21	-0.47	-0.22	4.07	0.08	-0.04	-0.69	2.18	-0.34	0.61	3.07	0.47	0.44
		(-9.47)	(-2.69)	(-0.71)	(13.24)	(0.33)	(-0.08)	(-1.88)	(1.44)	(-2.02)	(1.08)	(5.79)	(0.44)	
<i>Panel B: average parameter values, dependent variable is the level of institutional holdings</i>														
(1)	All firms	1.69	-4.37	-6.84	6.41	2.27	-1.85	-2.73	5.08	2.97	-1.08	1.96	-0.46	0.352
	firms	(0.49)	(-10.09)	(-2.83)	(3.06)	(5.77)	(-0.96)	(-7.23)	(3.46)	(10.23)	(-3.39)	(0.96)	(-0.55)	
(2)	Only small firms	-4.68	-3.79	-51.31	7.90	2.24	-3.15	-1.26	6.08	3.18	-0.08	3.94	-0.28	0.258
	large firms	(-1.39)	(-9.07)	(-2.57)	(11.83)	(7.51)	(-1.99)	(-1.59)	(3.80)	(7.83)	(-0.39)	(2.05)	(-0.47)	
(3)	Only large firms	26.81	-6.98	-3.55	2.97	2.95	1.13	-3.34	0.78	2.42	-2.61	7.74	0.83	0.155
		(12.05)	(-3.80)	(-1.80)	(2.00)	(2.93)	(0.46)	(-1.34)	(0.31)	(2.07)	(-1.06)	(3.90)	(0.41)	

Table 2: Analyst coverage and institutional holdings regressions.

Interpretation. Table 2 shows that the rural visibility deficit remains after controls. In Panel A, the rural dummy is negative for analyst coverage: all else equal, rural firms have about 0.32 fewer analysts in the full sample, with a statistically significant Fama–MacBeth t -statistic. In Panel B, the rural dummy is much larger for institutional ownership: institutions hold 4.37 percentage points less of rural firms' shares, and the effect reaches 6.98 percentage points for large firms. This table supports the attention and information channel directly: rural firms are followed and owned by fewer sophisticated market participants.

Table 4

Robustness tests of urban and rural turnover, 1983–2002.

In Panels A, B, and C, the dependent variable is the daily turnover for firm i in calendar day t . Panel D uses turnover/Return; while Panel E uses Return/volume as the dependent variable. For firms listed on NYSE/Amex, trading volume is doubled to allow for more meaningful comparison with firms on Nasdaq. The explanatory variables are a dummy variable equal to one if the firm is located in a rural locale (else zero), the market value and the natural logarithm of the market value, the number of analysts, employees, shareholders, inverse of stock price, dummy variables for manufacturing, banking, transportation, and mining industries (else zero), an S&P 500 Index dummy (else zero), and a dummy variable equal to one if the firm is listed on Nasdaq (else zero). The cross-sectional regressions weight each day equally. The parameter values are the average of the cross-sectional regressions. In parentheses are the Fama–MacBeth autocorrelation-adjusted t -statistics. Intercepts and coefficients on the four industry dummies are not reported to save space. The reported adjusted R^2 's are the time-series average of the daily values. The regression is as follows:

$$\begin{aligned} \text{Turnover}_i = & a_0 + a_1 \text{rural dummy}_i + a_2 (\text{market value}_i + e_1 \text{Log (market value)}_i) + a_3 \text{number of analysts}_i + a_4 \text{inverse of stock price}_i \\ & + a_5 \text{manufacturing dummy}_i + a_6 \text{bank dummy}_i + a_7 \text{transportation dummy}_i + a_8 \text{mining dummy}_i + a_9 \text{Log (number of employees)}_i \\ & + a_{10} (\text{number of shareholders})_i + a_{11} \text{S\&P 500 dummy}_i + a_{12} \text{Nasdaq dummy}_i + e_i \end{aligned}$$

Row	Time period	Rural dummy	Mkt value	Log (Mkt value)	No of analysts	1/Stock price	Log (Emply.)	Log (Shlders)	S&P 500 dummy	Nasdaq dummy	R^2_{adj}
<i>Panel A: subperiod results (all firms with stock price of \$5 or more)</i>											
(1)	1983–2002	-0.16	-0.18	0.09	0.03	-0.51	-0.06	-0.01	-0.12	0.08	0.057
	Obs = 5,049	(-10.86)	(-18.10)	(7.82)	(4.65)	(-2.23)	(-4.50)	(-0.97)	(-3.70)	(1.76)	
(2)	1983–1992	-0.10	-0.21	0.05	0.01	0.32	-0.02	0.01	-0.00	-0.02	0.041
	Obs = 2,529	(-22.66)	(-30.08)	(8.53)	(10.04)	(4.92)	(-3.87)	(1.68)	(-0.27)	(-0.73)	
(3)	1993–2002	-0.22	-0.16	0.13	0.04	1.35	-0.09	-0.02	-0.25	0.18	0.072
	Obs = 2,520	(-17.52)	(-9.41)	(11.49)	(6.02)	(-5.06)	(-5.85)	(-3.00)	(-9.55)	(5.94)	
<i>Panel B: subperiod results (all firms regardless of stock price)</i>											
(1)	1983–2002	-0.13	-0.19	0.10	0.03	0.03	-0.04	-0.01	-0.16	0.05	0.064
	Obs = 5,049	(-10.86)	(-18.10)	(8.01)	(4.57)	(2.70)	(-3.21)	(-1.41)	(-4.25)	(1.22)	
(2)	1983–1992	-0.08	-0.22	0.05	0.01	-0.01	-0.01	0.00	-0.03	-0.04	0.057
	Obs = 2,529	(-18.89)	(-26.51)	(9.97)	(9.93)	(-4.60)	(-3.27)	(1.18)	(-2.44)	(-2.13)	
(3)	1993–2002	-0.18	-0.16	0.14	0.04	0.06	-0.07	-0.01	-0.29	0.15	0.071

Table 4, part 1: Robustness tests for turnover.

Interpretation. The first part of Table 4 shows that the rural turnover gap is not an artifact of one sample period or one urban definition. Panel A finds a negative rural coefficient over 1983–2002 and in both subperiods, with the gap stronger in the price cutoff is relaxed, and Panel C shows it survives alternative urban definitions such as New York only, the five biggest cities, or all nonrural stocks. The interpretation is that geography is not merely proxying for one city, one period, or one sample screen.

Table 3

Average parameter values from daily cross-sectional regressions of percentage turnover, 1988–2002.

In the regressions, the dependent variable is the daily turnover for firm i on calendar day t . Turnover is defined as trading volume divided by shares outstanding. For firms listed on NYSE/Amex, trading volume is doubled to allow for more meaningful comparison with firms on Nasdaq. The fifteen explanatory variables are a dummy variable equal to one if the firm is located in a rural locale (else zero), the market value two days prior to calendar day t (shares outstanding times stock price), the natural logarithm of the market value, the number of analysts, employees, shareholders, percentage insider holdings, percentage institutional holdings, inverse of stock price, dummy variables for manufacturing, banking, transportation, and mining industries (else zero), a dummy variable equal to one if the firm is in the S&P 500 Index (else zero), and a dummy variable equal to one if the firm is listed on Nasdaq (else zero). The cross-sectional regressions weight each day equally. The parameter values are the average of the cross-sectional regressions. In parentheses are the Fama–MacBeth autocorrelation-adjusted t -statistics. Intercepts and coefficients on the four industry dummies are not reported to save space. The reported adjusted R^2 's are the time-series average of the daily cross-sectional R^2 's. There are 3,722 trading days during April 1988–December 2002. The regression is as follows:

$$\begin{aligned} \text{Turnover}_i = & a_0 + a_1 \text{Rural dummy}_i + a_2 (\text{market value}_i + e_1 \text{Log (market value)}_i) + a_3 \text{number of analysts}_i + a_4 \text{insider holdings}_i + a_5 \text{institutional holdings}_i \\ & + a_6 \text{inverse of stock price}_i + a_7 \text{manufacturing dummy}_i + a_8 \text{bank dummy}_i + a_9 \text{transportation dummy}_i + a_{10} \text{mining dummy}_i + a_{11} \text{Log (number of employees)}_i \\ & + a_{12} \text{Log (number of shareholders)}_i + a_{13} \text{S\&P 500 dummy}_i + a_{14} \text{Nasdaq dummy}_i + e_i \end{aligned}$$

Row	Sample	Rural dummy	Mkt value	Log (Mkt value)	No of analysts	Insider holdings	Instit. holdings	1/Stock price	Log (Emply.)	Log (Shlders)	S&P 500 dummy	Nasdaq dummy	R^2_{adj}
(1)	All firms	-0.17	-0.17	0.11	0.03	-0.26	0.31	-0.27	-0.08	-0.02	-0.19	0.16	0.074
	firms	(-11.04)	(-12.61)	(7.26)	(4.87)	(-8.10)	(6.71)	(-1.10)	(-6.21)	(-4.09)	(-6.17)	(4.45)	
(2)	Only Nasdaq firms	-0.22	-0.73	0.15	0.06	-0.26	0.09	-0.09	-0.12	-0.01	-0.32	0.17	0.110
	NYSE/Amex firms	(-10.72)	(-4.77)	(14.02)	(11.28)	(-6.08)	(1.31)	(-3.18)	(-8.61)	(-2.73)	(-3.86)		
(3)	Only NYSE/Amex firms	-0.07	-0.11	0.03	0.02	-0.23	0.40	0.04	-0.01	-0.02	-0.06		0.057
	Only small firms	(-8.55)	(-11.42)	(2.85)	(4.89)	(-11.82)	(10.67)	(3.71)	(-0.92)	(-2.22)	(-3.67)		
(4)	Only large firms	-0.19	-0.22	0.14	0.06	-0.30	-0.03	-0.64	-0.11	-0.00	-0.25		0.077
	Nasdaq	(-12.32)	(-0.26)	(9.12)	(12.46)	(-6.01)	(-0.40)	(-2.72)	(-7.75)	(-0.72)	(-3.35)		
(5)	Only large firms	-0.35	-0.22	0.01	0.05	-0.02	0.51	-4.73	-0.20	-0.01	-0.17		0.199
	Nasdaq	(-6.21)	(-2.70)	(0.24)	(9.12)	(-0.27)	(4.28)	(-4.91)	(-7.84)	(-0.54)	(-2.95)		
(6)	Only small firms	-0.07	-3.81	0.10	0.04	-0.29	0.03	0.54	-0.02	-0.00	0.19		0.050
	NYSE/Amex	(-5.60)	(-5.28)	(9.64)	(6.55)	(-12.65)	(0.09)	(3.83)	(-2.31)	(-0.47)	(2.80)		
(7)	Only small firms	-0.10	-0.03	-0.10	0.02	-0.15	0.41	1.12	0.02	-0.03	-0.02		0.082
	NYSE/Amex	(-0.81)	(-7.01)	(-8.50)	(5.97)	(-4.44)	(10.16)	(7.77)	(1.50)	(-3.11)	(-1.67)		

Table 3: Daily cross-sectional regressions of turnover, 1988–2002.

Interpretation. Table 3 is the central turnover regression. The rural dummy is negative and highly significant after controlling for market value, analysts, insider ownership, institutional ownership, price, employees, shareholders, S&P 500 membership, Nasdaq status, and industry. The effect is -0.17 percentage points of daily turnover for all firms, -0.22 for Nasdaq firms, and -0.07 for NYSE/Amex firms. Thus, rural firms trade less even conditional on standard determinants of trading. The larger Nasdaq coefficient foreshadows the paper's later market-maker explanation.

Panel C: results for different definitions of urban (all firms with stock price of \$5 or more)

(1)	New York only	-0.14	-0.11	0.06	0.02	-0.15	-0.03	-0.00	-0.07	-0.01	0.048
		(-7.81)	(-18.39)	(9.13)	(5.55)	(-0.90)	(-2.33)	(-0.96)	(-3.03)	(-0.82)	
(2)	Biggest 5 cities	-0.18	-0.18	0.10	0.03	-0.38	-0.06	-0.01	-0.15	0.08	0.062
		(-11.12)	(-15.55)	(9.49)	(4.50)	(-1.65)	(-4.73)	(-2.57)	(-4.96)	(1.76)	
(3)	All non-rural stocks	-0.12	-0.17	0.09	0.02	-0.41	-0.06	-0.02	-0.11	0.03	0.047
		(-12.56)	(-14.41)	(7.17)	(4.51)	(-1.84)	(-4.77)	(-3.07)	(-3.68)	(0.47)	

Panel D: using turnover/return (if return is not equal to zero) as a measure of liquidity

(1)	All firms	-0.68	-10.67	4.81	1.92	-289.86	-1.57	0.24	3.92	-4.70	0.062
		(-16.79)	(-27.17)	(11.71)	(7.76)	(-5.68)	(-6.36)	(1.28)	(3.94)	(-3.02)	
(2)	Nasdaq only	-11.02	-26.04	5.46	2.97	-218.64	-3.42	0.17	4.98		0.072
		(-10.16)	(-5.40)	(11.87)	(11.02)	(-5.73)	(-13.12)	(0.85)	(-2.37)		
(3)	NYSE/Amex only	-6.21	-9.46	1.70	1.42	-417.55	1.46	0.21	6.39		0.052
		(-8.14)	(-27.87)	(4.70)	(7.03)	(-5.24)	(5.74)	(0.54)	(6.42)		

Panel E: using return/trading volume (if trading volume is not equal to zero) as a measure of liquidity

(1)	All firms	0.14	0.45	-0.56	0.02	4.74	0.09	-0.00	0.69	0.51	0.042
		(7.67)	(12.53)	(-22.58)	(18.28)	(13.76)	(13.66)	(-0.23)	(25.10)	(9.91)	
(2)	Nasdaq only	0.22	9.19	-1.22	0.02	3.70	0.15	-0.04	0.24		0.044
		(6.17)	(3.63)	(-17.41)	(5.54)	(5.69)	(10.54)	(-2.80)	(2.99)		
(3)	NYSE/Amex only	0.01	0.12	-0.12	0.00	2.14	0.00	-0.02	0.12		0.112
		(3.62)	(4.90)	(-14.58)	(8.48)	(7.41)	(2.91)	(-0.09)	(20.93)		

Table 4, part 2: Alternative liquidity measures.

Interpretation. The second part of Table 4 tests whether the results depend on turnover as the liquidity measure. Panel D uses turnover divided by absolute returns; Panel E uses absolute return divided by trading volume as an illiquidity measure. The rural coefficients continue to point toward lower liquidity for rural stocks. This matters because simple turnover can be affected by volatility and price movements. The robustness checks show that rural firms look less liquid under both trading-quantity and price-impact style measures.

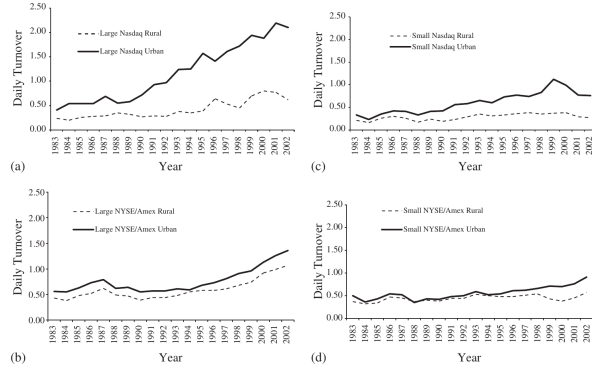


Figure 2: Annual turnover trends by size, exchange, and location.

Interpretation. Figure 2 makes the time-series pattern visible. For large Nasdaq firms, urban turnover rises dramatically from the early 1980s to 2002, while rural Nasdaq turnover increases much less. For large NYSE/Amex stocks, the rural and urban lines are much closer. The small-stock panels show a similar but less extreme pattern: urban Nasdaq turnover grows more than rural Nasdaq turnover, and NYSE/Amex gaps are smaller. The visual evidence is consistent with the regression evidence: the liquidity gap is especially tied to Nasdaq and grows as trading activity expands in the 1990s.

Table 6
Regressions of systematic and unsystematic turnover of urban and rural stocks on market capitalization and industry variables
Market turnover is obtained for each day over 1983–2002 by dividing the number of shares traded for all Nasdaq stocks by the total number of outstanding shares for all Nasdaq stocks. The NYSE/Amex market turnover is calculated in a similar manner. For each stock, the following time-series regression is run for each year:

$$\text{turnover}_{i,t} = \alpha_0 + \alpha_1 \text{turnover}_{m,t+1} + \alpha_2 \text{turnover}_{m,t} + \alpha_3 \text{turnover}_{m,t-1} + \epsilon_{i,t}$$

For each stock each year, α_2 is the same day market turnover coefficient. Systematic turnover is α_1 times the next day's market turnover, plus α_2 times the contemporaneous market turnover, plus α_3 times the previous day's market turnover. Unsystematic turnover is α_0 . For each year, cross-sectional regressions are run with the same day market turnover coefficient, the systematic turnover, and the unsystematic turnover as dependent variables. Explanatory variables include a dummy variable for a firm based in a rural location, capitalization, the inverse of capitalization, the natural logarithm of capitalization, industry dummies, employees, shareholders, and a S&P 500 Index dummy. Capitalization is measured at the beginning of each year. The Fama-MacBeth statistics are calculated using the time-series standard deviation of the cross-sectional regression t -statistics after adjusting for autocorrelation in the coefficients.

	Rural dummy	Mkt. value	1/Market value	LogMkt dummy	Mfg. dummy	Bank dummy	Transport dummy	Mining dummy	S&P dummy	Log (number empl.)	Log (number shrlhds)
<i>Panel A: Nasdaq stocks regressed on Nasdaq market turnover</i>											
<i>Same day mkt turnover coefficient (α_2):</i>											
Mean coefficient	-0.313	-0.542	3.484.0	0.690	-0.067	-0.332	-0.298		-0.534	-0.043	1.156
Median coefficient	-0.314	-0.034	1.150.5	0.351	0.070	-0.651	-0.276		-0.330	0.015	0.790
Fama-MacBeth t -stat	-7.12	-1.19	1.55	2.13	-0.63	-4.79	-4.49		-2.80	-0.62	2.90
<i>Systematic turnover ($\alpha_1 \text{ turnover}_{m,t+1} + \alpha_2 \text{ turnover}_{m,t} + \alpha_3 \text{ turnover}_{m,t-1}$):</i>											
Mean coefficient	-2.389	-2.236	4.714.6	2.863	-0.375	-5.408	-2.376		-3.186	0.219	4.299
Median coefficient	-1.942	-0.357	2.205.4	1.554	0.368	-3.341	-1.277		-2.619	0.293	3.094
Fama-MacBeth t -stat	3.61	1.41	0.43	3.43	0.50	4.55	1.45		1.92	0.73	3.52

Table 6, part 1: Systematic turnover regressions, Nasdaq panel.

Interpretation. Table 6 asks whether the systematic-turnover gap survives firm-level controls. In the Nasdaq panel, the rural dummy is negative for the same-day market-turnover coefficient and for systematic turnover. This means rural Nasdaq stocks respond less to marketwide trading even after controlling for market value, industry, employees, shareholders, and S&P 500 membership. The result strengthens the interpretation that rural location reduces the stock's role as a marketwide trading vehicle.

Table 5
Systematic and unsystematic components of firm turnover, 1983–2002

Market turnover is obtained for each day over 1983–2002 by dividing the number of shares traded for all Nasdaq stocks by the total number of outstanding shares for all Nasdaq stocks. The NYSE/Amex market turnover is calculated in a similar manner. For each stock each year, the following time-series regression is run using daily turnover:

$$\text{turnover}_{i,t} = \alpha_0 + \alpha_1 \text{turnover}_{m,t+1} + \alpha_2 \text{turnover}_{m,t} + \alpha_3 \text{turnover}_{m,t-1} + \epsilon_{i,t}$$

Systematic turnover is α_1 times market turnover the next day, plus α_2 times the contemporaneous market turnover, plus α_3 times the previous day's market turnover. Unsystematic turnover is α_0 . For each year average systematic turnover, average unsystematic turnover, average α_1 , and average α_2 are calculated for urban and rural stocks. T -statistics are calculated using the time-series standard deviation of the average coefficients after an adjustment for autocorrelation.

	Nasdaq stocks			NYSE/Amex stocks		
	Rural	Urban	Difference T -statistic	Rural	Urban	Difference T -statistic
Systematic Turnover	0.242%	0.623%	2.54	0.210%	0.271%	2.26
Unsystematic Turnover	0.035%	-0.056%	-2.17	0.032%	0.030%	-0.36
Coefficient Mkt. Turnover $t+1$	0.027	0.077	2.31	0.067	0.053	-0.69
Coefficient Mkt. Turnover t	0.392	0.878	13.34	0.702	0.917	4.61
Coefficient Mkt. Turnover $t-1$	0.008	0.056	4.48	0.045	0.052	0.34
Sum Coefficients $t-1$ to $t+1$	0.426	1.011	11.08	0.814	1.023	3.87
Average R^2	0.014	0.024		0.029	0.040	

Table 5: Systematic and unsystematic components of turnover.

Interpretation. Table 5 decomposes turnover into the component related to marketwide trading and an idiosyncratic component. The sharpest result is that rural Nasdaq stocks have much lower systematic turnover than urban Nasdaq stocks: 0.242% versus 0.623%. The contemporaneous market-turnover coefficient is also much lower for rural Nasdaq firms, 0.392 versus 0.878. This suggests that when investors trade because of broad market information or marketwide portfolio flows, they use urban stocks more intensively. Rural stocks are not just lower-volume stocks; they are less connected to marketwide trading activity.

	Unsystematic turnover (α_0)										
	Mean coefficient	Median coefficient	Fama-MacBeth t -stat	0.720	1.509	7,189.5	-0.265	0.314	1.638	2.971	-0.145
<i>Panel B: NYSE/Amex stocks regressed on NYSE/Amex market turnover</i>											
<i>Same day Mkt turnover coefficient (α_2):</i>											
Mean coefficient	-0.135	-0.184	1,949.0	0.174	-0.172	-0.292	-0.351		-0.151	0.139	0.003
Median coefficient	-0.138	-0.186	97.8	0.139	-0.146	-0.320	-0.358		-0.155	0.174	0.017
Fama-MacBeth t -stat	-5.72	-6.62	1.08	3.75	-6.46	-5.00	-7.26		-3.29	1.41	0.22
<i>Systematic turnover ($\alpha_1 \text{ turnover}_{m,t+1} + \alpha_2 \text{ turnover}_{m,t} + \alpha_3 \text{ turnover}_{m,t-1}$):</i>											
Mean coefficient	-0.457	-0.441	7,154.8	0.430	-0.385	-0.749	-0.888		-0.612	0.167	-0.076
Median coefficient	-0.273	-0.384	835.3	0.367	-0.407	-0.656	-0.800		-0.617	0.324	-0.024
Fama-MacBeth t -stat	-1.64	-7.99	1.20	2.42	-5.17	-3.79	-4.43		-2.72	1.09	-1.10
<i>Unsystematic turnover (α_0):</i>											
Mean coefficient	0.052	-0.019	-5,510.6	-0.085	-0.083	-0.029	0.098		0.546	-0.219	0.054
Median coefficient	0.060	-0.033	-721.3	-0.139	-0.150	-0.159	0.100		0.282	-0.148	0.012
Fama-MacBeth t -stat	0.43	-0.50	-1.24	-0.96	-0.92	-0.17	0.44		2.07	-1.49	0.91

Table 6, part 2: Continuation for unsystematic and NYSE/Amex turnover.

Interpretation. The continuation shows the NYSE/Amex panel and the unsystematic turnover estimates. The rural dummy remains negative for the same-day market-turnover coefficient among NYSE/Amex stocks, but the magnitude is much smaller than on Nasdaq. The unsystematic components are less central to the paper's argument than the systematic coefficients. Overall, Table 6 indicates that geography affects the systematic part of liquidity, and that the effect is most pronounced where market making and investor visibility are more decentralized.

Table 7

Average parameter values from monthly cross-sectional regressions of relative spreads on a rural dummy, market value, log of market value, number of analysts, inverse of stock price, and four industry dummies, 1993–1995.

In the regressions, the dependent variable is the average quoted or effective relative spread for firm i in month j . Relative spreads for each trade are calculated by dividing the quoted (effective) spread by the contemporaneous average of the bid and ask prices. The nine explanatory variables are a dummy variable equal to one if the firm is located in a rural location (else zero), the market value two days prior to calendar month j (shares outstanding times stock price), the natural logarithm of the market value, the number of analysts covering the stock, the reciprocal of the mean price during the month, and dummy variables for manufacturing, banking, transportation, and mining industries (else zero). Small firms are defined as firms with a market capitalization less than the median NYSE firm in a given year while large firms are firms with a market value equal or greater than the median NYSE firm. The cross-sectional regressions weight each month equally. The parameter values are the average of the cross-sectional regressions. In parentheses are the Fama-MacBeth autocorrelation-adjusted t -statistics. There are 36 months during January 1993–December 1995. The regression is as follows:

$$\text{Spread}_{ij} = \alpha_0 + \alpha_1 \text{rural dummy}_{ij} + \alpha_2 \text{market value}_{ij} + \alpha_3 \text{Log}(\text{market value})_{ij} + \alpha_4 \text{number analysts}_{ij} + \alpha_5 (1/\text{price})_{ij} \\ + \alpha_6 \text{manufacturing dummy}_{ij} + \alpha_7 \text{bank dummy}_{ij} + \alpha_8 \text{transportation dummy}_{ij} + \alpha_9 \text{mining dummy}_{ij} + \epsilon_{ij}.$$

Row	Relative spread measure	Intercept	Rural dummy	Mkt value	Log (Mkt value)	Number analysts	1/Price dummy	Mfg. dummy	Bank dummy	Transp. dummy	Mining dummy
R^2_{adj}											
<i>Panel A: Nasdaq Stocks</i>											
(1)	Quoted spread	23.08 (11.70)	0.49 (12.89)	4.21 (4.47)	−1.64 (−10.59)	−0.03 (−4.52)	5.49 (3.38)	−0.04 (−1.11)	0.30 (1.64)	0.08 (0.94)	0.22 (2.21)
(2)	Effective spread	18.43 (12.03)	0.34 (7.97)	3.32 (5.54)	−1.30 (−20.49)	−0.03 (−6.31)	7.79 (9.56)	−0.05 (−1.75)	0.04 (0.37)	0.06 (0.97)	0.14 (1.35)
(3)	Effective spread	14.76 (11.70)	0.27 (12.89)	2.79 (4.47)	−1.06 (−10.59)	−0.02 (−4.52)	8.09 (3.38)	−0.01 (−1.11)	0.09 (1.64)	0.09 (0.94)	0.381 (2.21)

Table 7, part 1: Relative spread regressions, Nasdaq panel.

Interpretation. Table 7 directly examines trading costs during 1993–1995. In the Nasdaq panel, the rural dummy is positive and statistically significant for quoted spreads and effective spreads across trade-size categories. For example, the quoted spread coefficient on the rural dummy is 0.49, and the effective-spread coefficients are also positive. This means that, conditional on size, analyst coverage, price, and industry, rural Nasdaq stocks are more expensive to trade. The table connects lower visibility to an economically concrete cost faced by investors.

Table 8

Determinants of the number of market makers for Nasdaq stocks, May–December 1995

For each month the number of market makers, number of large market makers, and Herfindahl index for each Nasdaq stock is regressed on a dummy variable for rural location and other explanatory variables. Number of market makers is the number that reported any volume as a market maker for the month. Small and large firms are smaller/larger than the median capitalization at the month's beginning. Large market makers made markets in at least 200 stocks during the month. The Herfindahl Index for each stock is calculated by summing the squared percentage of shares traded by each market maker. Market value is the product of shares outstanding and price at the month's start. Volatility is the mean daily squared return for the prior 12 months. Volume is the mean daily volume for the prior twelve months. Return is the cumulative return over the prior twelve months. The rural dummy takes a value of one if the company headquarters is less than 100 miles from the nearest metropolitan area of one million or more. Urban firms have headquarters within one of the ten largest metropolitan areas of the U.S. Other firms are omitted. The average coefficients across the eight months are reported in the table with average t -statistics in parentheses. Finally, R^2 is the average R^2 across the eight regressions.

	Dependent variables		
	Number of market makers	Number of large market makers	Herfindahl index
Intercept	−15.81 (−5.31)	−15.74 (−6.80)	0.80 (13.29)
Rural dummy	−2.34 (−7.22)	−1.61 (−6.94)	0.05 (4.61)
Log(market value)	2.38 (10.29)	2.12 (11.97)	−0.05 (−10.57)
Volatility	34.67 (2.57)	28.48 (2.60)	−0.74 (−2.62)
Volume	11.87 (2.18)	8.21 (2.08)	0.01 (0.60)
Return	−1.10 (−4.22)	−0.78 (−5.15)	0.01 (0.95)
Manufacturing dummy	0.44 (1.23)	0.41 (1.60)	−0.02 (−1.90)
Banking dummy	−1.93 (−4.48)	−1.86 (−5.82)	0.04 (3.44)
Transportation dummy	0.31 (0.36)	0.12 (0.24)	0.01 (0.71)
Mining dummy	1.39 (1.99)	0.85 (1.86)	−0.01 (−0.31)
R^2	0.332	0.413	0.168
Observations	1,852	1,852	1,852

Table 8: Determinants of Nasdaq market makers.

Interpretation. Table 8 explains why rural Nasdaq firms face higher costs. Rural Nasdaq stocks have about 2.34 fewer market makers and 1.61 fewer large market makers after controls, and their Herfindahl index is higher. Fewer market makers and higher concentration mean weaker dealer competition. This table therefore provides a mechanism linking rural headquarters to higher spreads: remote firms attract less market-making attention, making their trading environment thinner and more concentrated.

(4)	Effective spread > 1,000 shares	12.96 (26.68)	0.20 (8.25)	2.59 (6.17)	−0.93 (−23.60)	−0.02 (−6.28)	8.33 (15.01)	−0.02 (−0.95)	0.11 (1.33)	0.10 (2.24)	−0.05 (−1.00)	0.369
<i>Panel B: NYSE/Amex Stocks</i>												
(1)	Quoted spread	2.42 (23.02)	0.01 (0.36)	0.12 (5.89)	−0.15 (−20.22)	−0.01 (−1.04)	13.10 (22.87)	0.02 (0.49)	0.04 (0.42)	−0.04 (−0.34)	−0.01 (−0.29)	0.868
(2)	Effective spread < 500 shares	1.86 (9.56)	0.01 (0.91)	0.11 (16.15)	−0.12 (−9.67)	−0.00 (−0.69)	9.25 (36.50)	0.00 (0.40)	0.04 (4.95)	−0.01 (−1.29)	0.00 (0.08)	0.804
(3)	Effective spread 500–1,000 shares	2.87 (15.54)	−0.02 (−8.16)	0.14 (10.76)	−0.20 (−16.63)	0.00 (9.41)	8.69 (54.34)	0.00 (1.02)	0.06 (9.23)	−0.02 (−3.48)	−0.03 (−2.29)	0.820
(4)	Effective spread > 1,000 shares	3.00 (20.85)	−0.03 (−7.37)	0.15 (11.66)	−0.21 (−19.99)	0.00 (5.29)	9.60 (54.57)	0.01 (2.62)	0.06 (5.93)	−0.01 (−1.08)	−0.04 (−2.70)	0.785

Table 7, part 2: Continuation and NYSE/Amex panel.

Interpretation. The continuation shows that the rural spread effect is much weaker on NYSE/Amex. Rural coefficients for listed stocks are small and often statistically insignificant or negative for effective spreads. This contrast is important: geography matters most in the dealer-based Nasdaq environment. The evidence points to a microstructure channel rather than a universal mechanical relation between location and spreads. Listed exchange structure appears to mitigate some of the disadvantages of being rural.

Table 9

Changes in turnover around listing on the NYSE or Amex from Nasdaq, 1983–2002

Average daily turnover is calculated for urban and rural stocks moving from Nasdaq to the NYSE or Amex using either six months or one year before and after listing. The turnover difference is the mean daily turnover after listing minus the mean daily turnover before. The rural dummy takes a value of one for companies located at least 100 miles from any metropolitan area of one million or more people. The market capitalization is computed on the first day the stock trades on the NYSE or Amex. The difference in volatilities is the standard deviation of daily stock returns after listing minus the standard deviation of daily returns before. The market turnover difference is the mean daily turnover of the equal-weighted portfolio of all listed and Nasdaq stocks after the stock lists minus the mean daily turnover before. The regression is as follows:

$$\text{Turnover Diff}_i = \alpha_0 + \alpha_1 \text{rural}_i + \alpha_2 \text{Log}(\text{Mkt cap})_i \\ + \alpha_3 \text{Return before}_i + \alpha_4 \text{Return after}_i + \alpha_5 \text{Diff}_i + \alpha_6 \text{turnover diff}_{6M} + \epsilon_i.$$

Row	Intercept	Rural dummy	Log (Mkt value)	Mean daily return before	Mean daily return after	Difference in volatility	Market turnover difference	R^2
<i>Six months before and after listing</i>								
(1)	−2.65 (−13.92)	1.25 (3.40)						0.012
(2)	5.61 (5.20)	1.42 (3.94)	−1.59 (−7.23)	−102.6 (−1.70)	185.0 (2.59)			0.115
(3)	5.69 (5.19)	1.44 (3.97)	−1.61 (−7.20)	−100.0 (−1.63)	187.4 (2.53)	27.21 (0.38)	0.02 (0.05)	0.115
<i>One year before and after listing</i>								
(4)	−2.60 (−11.70)	1.29 (3.19)						0.011
(5)	4.10 (3.66)	1.34 (3.37)	−1.33 (−5.82)	12.18 (0.18)	297.2 (3.50)			0.069
(6)	4.12 (3.62)	1.34 (3.31)	−1.32 (−5.69)	11.91 (0.18)	300.6 (3.39)	8.10 (0.17)	−0.21 (−0.46)	0.069

Table 9: Turnover changes after moving from Nasdaq to NYSE/Amex.

Interpretation. Table 9 provides an event-style test. When firms move from Nasdaq to NYSE or Amex, rural firms experience a larger increase in turnover relative to urban firms. The rural dummy is positive and significant over both six-month and one-year windows. This supports the visibility interpretation: listed-exchange status seems to make rural firms more noticeable and easier to trade. The result also helps explain why rural-urban gaps are smaller on NYSE/Amex than on Nasdaq.

6 Interpretation and economic logic

6.1 Why location can matter in a national stock market

The paper’s key insight is that national trading technology does not eliminate local information and familiarity frictions. Investors can technically trade any listed stock, but they do not pay equal attention to all stocks. Urban firms are located near more investors, brokers, analysts, financial media, and institutional money managers. This proximity makes urban firms more familiar and lowers informal information costs.

6.2 Why the Nasdaq effect is stronger

Nasdaq liquidity during the sample period depends heavily on market makers. If rural firms attract fewer investors and fewer intermediaries, market makers have weaker incentives to provide liquidity, and dealer competition is thinner. This raises spreads and lowers turnover. NYSE/Amex listing, by contrast, provides a centralized specialist structure and a visibility benefit that can partially compensate for rural location.

6.3 Why systematic turnover matters

The distinction between systematic and unsystematic turnover is important because it shows that urban stocks are not just trading more due to firm-specific events. They also trade more when the whole market trades. This suggests that investors choose urban stocks as more natural vehicles for broad portfolio rebalancing, marketwide news, or liquidity needs. Rural stocks are less integrated into this marketwide trading network.

7 Critical evaluation

7.1 Strengths

- The paper uses many liquidity proxies rather than relying only on turnover.
- The rural and urban definitions are intentionally sharp, reducing ambiguity in location classification.
- The results survive extensive controls for firm size, exchange, industry, analyst coverage, price, S&P 500 membership, and ownership.
- The paper links reduced liquidity to a plausible mechanism: lower analyst/institutional attention and thinner Nasdaq market making.
- The Nasdaq-to-NYSE/Amex listing test gives additional support to the visibility channel.

7.2 Limitations

- Headquarters location is an imperfect proxy for where investors obtain information or where real operations are located.
- Location is not randomly assigned; rural firms may differ in unobserved ways from urban firms.

- The results are strongest in the 1983–2002 and 1993–1995 market structure, so the magnitude may differ after decimalization, electronic trading, and the rise of ETFs.
- The classification excludes firms that are neither clearly rural nor in the top urban centers, which improves contrast but leaves a middle category unanalyzed.
- The evidence is mainly cross-sectional; the listing-event test helps but is not a fully randomized experiment.

8 Takeaways for future research

- Geography can proxy for investor recognition and information access, even in centralized financial markets.
- Liquidity should be analyzed as both trading quantity and trading cost; the mechanisms can differ.
- Market structure matters: the same geographic friction has a different effect on Nasdaq versus NYSE/Amex.
- Location-based information frictions may interact with analyst coverage, institutional ownership, media coverage, and local investor bases.
- Similar methods could be used to study modern settings, such as whether remote firms still suffer liquidity disadvantages after electronic trading, online disclosure, and Reg FD.

9 Bottom line

Loughran and Schultz show that corporate geography is a liquidity-relevant firm characteristic. Rural firms are not merely smaller or industry-different; after a broad set of controls, they are followed by fewer analysts, owned by fewer institutions, traded less, less tied to marketwide trading, and more costly to trade on Nasdaq. The paper’s main contribution is to demonstrate that information access and familiarity are not abstract concepts: they show up in measurable market liquidity.